

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
10	U04						*****VALVES*****			810	NIPPLE TO BE CUT FROM SCH 160 PIPE.			
20										B1	ALL BOLTS ARE METRIC DIA.			
20	U04	VA5					***** MODULAR VALVES *****			B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10			
30										B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22			
30	U04		MODULAR DBB	M03AM10	0.5	3	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR	0	0					
40	U04		MODULAR DBB	K03AM10	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	0	0					
50	U04	VAV	MODULAR DBB	K03AM10	3	3	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	0	0					
60	U04	VAW	MODULAR DBB	M03EM10	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.			
70	U04	VAW	MODULAR DBB	M03EM10	1	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.			
80										SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.			
80	U04	VB2					***** BALL VALVES *****							
90														
90	U04		BALL	B03AM10	0.75	2	BALL 300# RF INC 825 INC 625 TRIM FLOATING METAL FB LO	0	0					
100	U04	VA5	BALL	B03AM10	3	4	BALL 300# RF INC 825 INC 625 TRIM FLOATING METAL FB LO	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.			
110	U04		BALL	A03AM10	3	3	BALL 300# RF INC 825 INC TRIM 625 FLOATING METAL RB LO	0	0					
120	U04		BALL	A03AR10	4	4	BALL 300# RF LTCS + CLAD INC 825 INC 625 TRIM FLOATING METAL RB LO	0	0					
130	U04		BALL	A03AR16	6	8	BALL 300# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION METAL RB GO	0	0					
140	U04	VA5	BALL	B03AR12	6	6	BALL 300# RF LTCS + CLAD INC 825 INC TRIM 625 FLOATING METAL FB GO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS			
150	U04	VA5	BALL	B03AR16	8	12	BALL 300# RF LTCS + CLAD INC 825 INC TRIM 625 TRUNNION METAL FB GO	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.			
160										VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.			
160	U04						***** BUTTERFLY VALVES *****							
170														
170	U04		BUTTERFLY	T03AR63	10	12	BUTTERFLY 300# RF LTCS + CLAD INC 825 INC TRIM 625 TOV METAL DBL FLG SHORT PAT GO	0	0					
180														
180	U04	VQ2					***** GATE VALVES *****			VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.			
190														
190	U04		GATE	G03AM49	0.75	3	GATE 300# RF INC 825 INC 625 TRIM WEDGE HW	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.			
200														
200	U04						***** GLOBE VALVES *****							
210														
210	U04		Globe	L03AM49	0.75	3	GLOBE 300# RF INC 825 INC 625 TRIM OS&Y HW	0	0	VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.			
220	U04		Globe	L03AR44	8	10	GLOBE 300# RF LTCS + 825 CLAD INC 625 TRIM OS&Y GO	0	0					
230	U04		Globe	L03AR49	4	6	GLOBE 300# RF LTCS + CLAD INC 825 INC TRIM 625 OS&Y HW	0	0	VQ2	QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS.			
240														
240	U04						***** CHECK VALVES *****			53	USE OF NPS 1/2 IS RESTRICTED TO ORIFICE FLANGE TAP CONNECTIONS.			
250														
250	U04	VC1	Check	C03AM51	0.75	1.5	CHECK PISTON 300# RF INC 825 TRIM 625	0	0	381	ALL ALLOY 825 MATERIALS SHOULD FOLLOW REQUIREMENTS IN CPM-SU-5013. ALL ALLOY 825 MATERIALS SHALL BE STABILIZED ANNEALED.			
260	U04	VC2	Check	C03AR53	12	12	CHECK DUAL DBL FLG 300# RF LTCS + CLAD INC 825 TRIM 625	0	0					
270	U04	VC2	Check	C03AM46	2	3	CHECK DUAL WAFER LUG 300# RF INC 825 INC TRIM 625 THROU BOLTED	0	0					
280	U04	VC2	Check	C03AR46	4	10	CHECK DUAL WAFER LUG 300# RF LTCS + CLAD INC 825 TRIM INC 625 THROU BOLTED	0	0					
290														
290	U04						***** PIPE *****							
300														
300	U04	810	PIPE	5361608-CD	0.5	0.75	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 COLD DRAWN ANNEALED PE	0	0					
310	U04		PIPE	5361603-CD	2	6	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 COLD DRAWN ANNEALED BE	0	0					
320	U04		PIPE	5732002-CD	1.5	1.5	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 COLD DRAWN ANNEALED PE	0	0					
330	U04		PIPE	5361603-WLD	8	12	PIPE SCH 40S WLD ALLOY 825 B474 UNS N08825 CL 3 STABILIZED ANNEALED BE	0	0					
340	U04		PIPE	5732001-CD	0.75	1	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 COLD DRAWN ANNEALED PE	0	0					
350														
350	U04						***** FITTINGS *****							
360														
360	U04		Cap	5424219	1.5	12	CAP SCH 40S ALLOY 825 B366-WPNICMC-S	1.5	2					
370	U04		Cap	5424270	0.75	1	CAP SCH 80S ALLOY 825 B366-WPNICMC-S	1	1					
380														
380	U04		ELBOW	5424727	1.5	6	ELL 45 DEG LR SCH 40S ALLOY 825 B366-WPNICMC-S	0	0					
390	U04		ELBOW	5424727-WLD	8	12	ELL 45 DEG LR SCH 40S ALLOY 825 B366-WPNICMC-WX	0	0					
400														
400	U04		RED CONC	5424733	1.5	4	REDUCER CONC SCH 40S ALLOY 825 B366-WPNICMC-S	0	0					
410	U04		RED CONC	5424733-WLD	4	10	REDUCER CONC SCH 40S ALLOY 825 B366-WPNICMC-WX	0	0					
420	U04		RED CONC	5425143	0.75	1	REDUCER CONC SCH 40S/80S ALLOY 825 B366-WPNICMC-S	0	0					
430	U04		RED CONC	5425122	0.75	0.75	REDUCER CONC SCH 80S ALLOY 825 B366-WPNICMC-S	8	12					
440	U04		RED ECC	5424734	1.5	4	REDUCER ECC SCH 40S ALLOY 825 B366-WPNICMC-S	2	6					
450	U04		RED ECC	5424734-WLD	4	10	REDUCER ECC SCH 40S ALLOY 825 B366-WPNICMC-WX	8	12					
460	U04		RED ECC	5425145	0.75	1	REDUCER ECC SCH 40S/80S ALLOY 825 B366-WPNICMC-S	1.5	2					
470	U04		RED ECC	5425123	0.75	0.75	REDUCER ECC SCH 80S ALLOY 825 B366-WPNICMC-S	1	1					
480														
480	U04		TEE RED	5424724	1.5	4	TEE RED SCH 40S ALLOY 825 B366-WPNICMC-S	0	0					
490	U04		TEE RED	5424724-WLD	4	10	TEE RED SCH 40S ALLOY 825 B366-WPNICMC-WX	0	0					
500	U04		TEE RED	5425124	0.75	1	TEE RED SCH 40S/80S ALLOY 825 B366-WPNICMC-S	0	0					
510	U04		TEE RED	5425126	0.75	0.75	TEE RED SCH 80S ALLOY 825 B366-WPNICMC-S	0	0					
			CONSTRUCTION		CORR.ALLOW.		SERVICE	MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS
			ALL SIZES - BUTT WELD SEAL WELDING - NONE PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 1 MM		SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 3.8 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: ALLOY 825 FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 230 °C		12/16/2024	BY OZAC	U04	Page 1 of 2	300A80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
520	U04		Tee	5424725	1.5	6	TEE SCH 40S ALLOY 825 B366-WPNICMC-S	0	0	
530	U04		Tee	5424725-WLD	8	12	TEE SCH 40S ALLOY 825 B366-WPNICMC-WX	0	0	
540	U04		Tee	5425119	0.75	1	TEE SCH 80S ALLOY 825 B366-WPNICMC-S	0	0	
550										
550	U04		ELBOW	5425113	0.75	1	ELL 45 DEG LR SCH 80S ALLOY 825 B366-WPNICMC-S	0	0	
560	U04		ELBOW	5424722	1.5	6	ELL 90 DEG LR SCH 40S ALLOY 825 B366-WPNICMC-S	0	0	
570	U04		ELBOW	5424722-WLD	8	12	ELL 90 DEG LR SCH 40S ALLOY 825 B366-WPNICMC-WX	0	0	
580	U04		ELBOW	5425111	0.75	1	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPNICMC-S	0	0	
590										
590	U04						***** OLETS *****			
600										
600	U04		OLETS	5513495	1.5	2	NIPOFLANGE 300# RF SCH 40S ALLOY 825 B564 UNS N08825	8	12	
610	U04		OLETS	5513479-40160	1.5	1.5	NIPOFLANGE 300# RF SCH 40S/160 ALLOY 825 B564 UNS N08825	0	0	
620	U04		OLETS	5513491	0.75	1	NIPOFLANGE 300# RF SCH 40S/80S ALLOY 825 B564 UNS N08825	2	6	
630	U04		OLETS	5504519	1.5	1.5	NIPOFLANGE FLAT BOTTOM 300# RF SCH 40S ALLOY 825 B564 UNS N08825	2	12	
640	U04		OLETS	5504518	0.75	1	NIPOFLANGE FLAT BOTTOM 300# RF SCH 80S ALLOY 825 B564 UNS N08825	2	12	
650										
650	U04	UJ2	ELBOFLG	5516626-40160	1.5	1.5	ELBOFLANGE 300# RF SCH 40S/160 ALLOY 825 B564 UNS N08825	4	12	
660										
660	U04	UJ2	OLETS	5513437	0.75	1	WELDOLET SCH 40S/80S ALLOY 825 B564 UNS N08825	4	12	
670										
670	U04	FS1					***** FLANGES *****			
680										
680	U04		Flange	5569290	0.5	1	FLG WN 300# RF ALLOY 825 SCH 80S B564 UNS N08825	0	0	
690	U04		Flange	5569276	1.5	12	FLG WN 300# RF ALLOY 825 SCH 40S B564 UNS N08825	0	0	
700	U04	FA8	FLG RED	5614310	0.5	0.5	FLG SCR D (FNPT) REDUCING 300# RF ALLOY 825 B564 UNS N08825	0.75	0.75	
710	U04		Blind	5617159	0.5	12	FLG BLIND 300# RF ALLOY 825 B564 UNS N08825	0	0	
720										
720	U04	53, 381					***** ORIFICE FLANGES *****			
730										
730	U04		ORIFICE FLG	5650830	2	12	WN ORIF SET 300# RF ALLOY 825 B564 UNS N08825 SCH 40S	0	0	
740										
740	U04						***** NIPPLES *****			
750										
750	U04						NIPPLE CUT FROM PIPE			
760										
760	U04						***** GASKETS *****			
770										
770	U04		GASKET	5673657	0.5	12	GASKET 300# ALLOY 825 SPIRAL WOUND GRAPHITE IR	0	0	
780										
780	U04						***** BOLTING *****			
790										
790	U04	B1	BOLTING	5675770	0.5	6	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
800										
800	U04	SB6	BOLTING	5675675	12	12	JACK SCREW HEX HEAD A320 GR L7M	0	0	
810										
810	U04						***** BLINDS & SPACERS *****			
820										
820	U04		PADDLE	5750231-300	12	12	PADDLE BLANK 300# RF ALLOY 825 B424 UNS N08825	0	0	
830	U04		PADDLE	5750232-300	12	12	PADDLE SPACER 300# RF ALLOY 825 B424 UNS N08825	0	0	
840										
840	U04		SPEC BLIND	5749683	0.75	10	SPECTACLE BLIND 300# RF ALLOY 825 B424 UNS N08825	0	0	
850										
860										
860	U04						THREAD SEALANT-----PTFE PASTE			

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - NONE PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 3.8 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: ALLOY 825 FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 230 °C	12/16/2024	BY OZAC	U04	Page 2 of 2	300A80